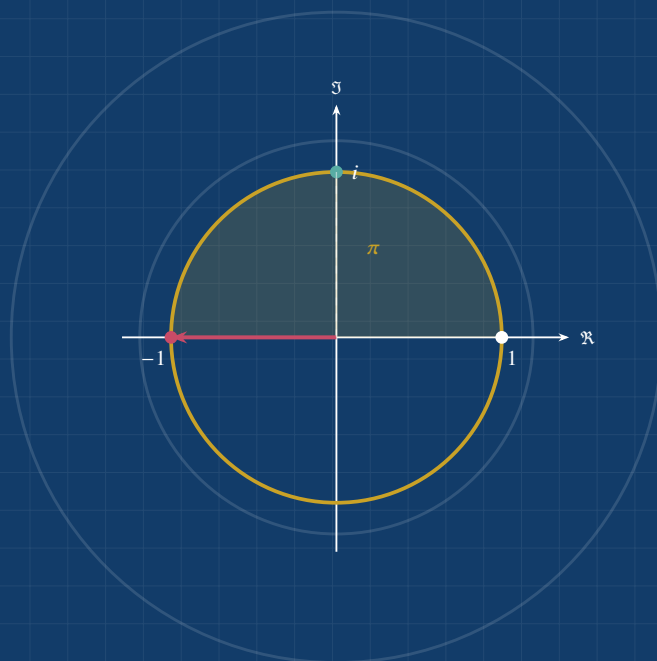


$$e^{\pi i} + 1 = 0$$

The Cognitive Structure of Classical Mathematics

A step-by-step analysis of **Part VI** of
Where Mathematics Comes From
by George Lakoff & Rafael E. Núñez (2000/2001)

Case Studies 1–4 · book pp. 381–452 · pdf pp. 400–471



CS 1

Analytic Geometry
& Trigonometry

CS 2

What is e ?

CS 3

What is i ?

CS 4

How the ideas
fit together

Gentlemen, that is surely true, it is absolutely paradoxical; we cannot understand it, and we don't know what it means. But we have proved it, and therefore we know it must be truth.

BENJAMIN PEIRCE, lecturing on Euler's proof that $e^{\pi i} = -1$ — quoted by Lakoff & Núñez, book p. 383 / pdf p. 402.

The whole of Part VI is an attack on that sentence.

What is in this dossier

■ the four case studies, reconstructed step by step

■ every metaphor and blend drawn as a diagram

■ the derivations, with the load-bearing step marked

■ verbatim evidence with page citations

■ a candid audit: six places the argument leaks

■ a reading plan against your own shelf

Contents

$$e^{\pi i} + 1 = 0$$

Every section carries book- and pdf-page citations.

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How to read the apparatus

PURPLE SMALL CAPS = a metaphor · BLUE SMALL CAPS = a blend · gold rule = verbatim quotation · (book p. N/pdf p. M) = both paginations

teal idea worth keeping

blue key result

amber caution

red criticism or demolition

1 Orientation: what Part VI is actually for

Parts I–V of *Where Mathematics Comes From* build an apparatus. **Part VI is the stress test.** Lakoff & Núñez take the single most revered formula in classical mathematics and ask whether their method — *mathematical idea analysis* — can say something about it that axioms, definitions, theorems and proofs cannot.

The opening move is rhetorical but surgical: Benjamin Peirce, one of Harvard's leading nineteenth/century mathematicians, lecturing on Euler's proof and confessing that he had proved something he did not understand. Part VI is a sustained argument that **proof is not understanding**, and that understanding has a describable structure built from three mechanisms the book has already introduced.

Grounding metaphors

Map *bodily, pre-mathematical* experience onto mathematics.

e.g. **ARITHMETIC IS OBJECT CONSTRUCTION**

Linking metaphors

Map one *mathematical* domain onto another, exporting inference.

e.g. **FUNCTIONS ARE NUMBERS**

Conceptual blends

Fuse two domains into a third with *emergent* entities and entailments.

e.g. the **UNIT CIRCLE BLEND**

plus the **BMI**, the Basic Metaphor of Infinity, wherever a limit appears

1.1 The standard of success they set themselves

This is worth copying into the front of your notebook, because it is the yardstick against which the rest of the Part must be measured.

Requirements for an adequate mathematical idea analysis (book p. 436 / pdf p. 455)

Of an equation, definition, axiom or theorem, an idea analysis must give:

1. the **metaphor-and-blend structure** that makes it true *relative to that structure*;
2. the **conceptual relationships** among the elements that appear in, or are presupposed by, it;
3. the **ideas expressed by the symbols**;
4. the **grounding** of the concepts used in the analysis.

The consequence they draw immediately is the most philosophically loaded sentence in Part VI:

“The “truth” of an equation, axiom, or theorem depends on what it means. That is, truth is relative to a conceptual structure. . . Thus, $e^{\pi i} + 1 = 0$ is not true of Euclidean geometry or the arithmetic of real numbers for an obvious reason: Those mathematical domains do not have the right idea structure to give meaning to the equation, much less to make it true.

BOOK P. 436 / PDF P. 455

I return to this claim — and to the two very different readings it admits — in §6.

1.2 Why this particular equation

Their own justification, stated on the Part-opening page, is that Euler's equation is a *junction*: it is the one place where every branch of classical mathematics is forced into contact.

“It brings together in one equation the most important numbers in mathematics: e , π , i , 1, and 0. In doing so, this one equation implicitly states a relationship among the branches of classical mathematics—arithmetic, algebra, geometry, analytic geometry, trigonometry, calculus, and complex variables.

BOOK P. 381 / PDF P. 400

And their statement of the problem is refreshingly blunt: the usual gloss on exponentiation simply *cannot* be what is going on.

The problem, stated precisely (book p. 384 / pdf p. 403)

$e^{\pi i}$ makes no sense as “ e multiplied by itself π times, and then i times.” π is an infinite decimal; $i = \sqrt{-1}$. **“What could ‘ $\sqrt{-1}$ times’ possibly mean?”** A different answer is required — one that follows from the *conceptual content* of e , π , i and -1 .

1.3 The whole architecture on one page

Everything Part VI does is an assembly. Below is the full dependency graph: bodily experience at the foot, the equation at the crown. Read it upward. Each arrow is a metaphor or a blend that *exports inferential structure* from the level below to the level above.

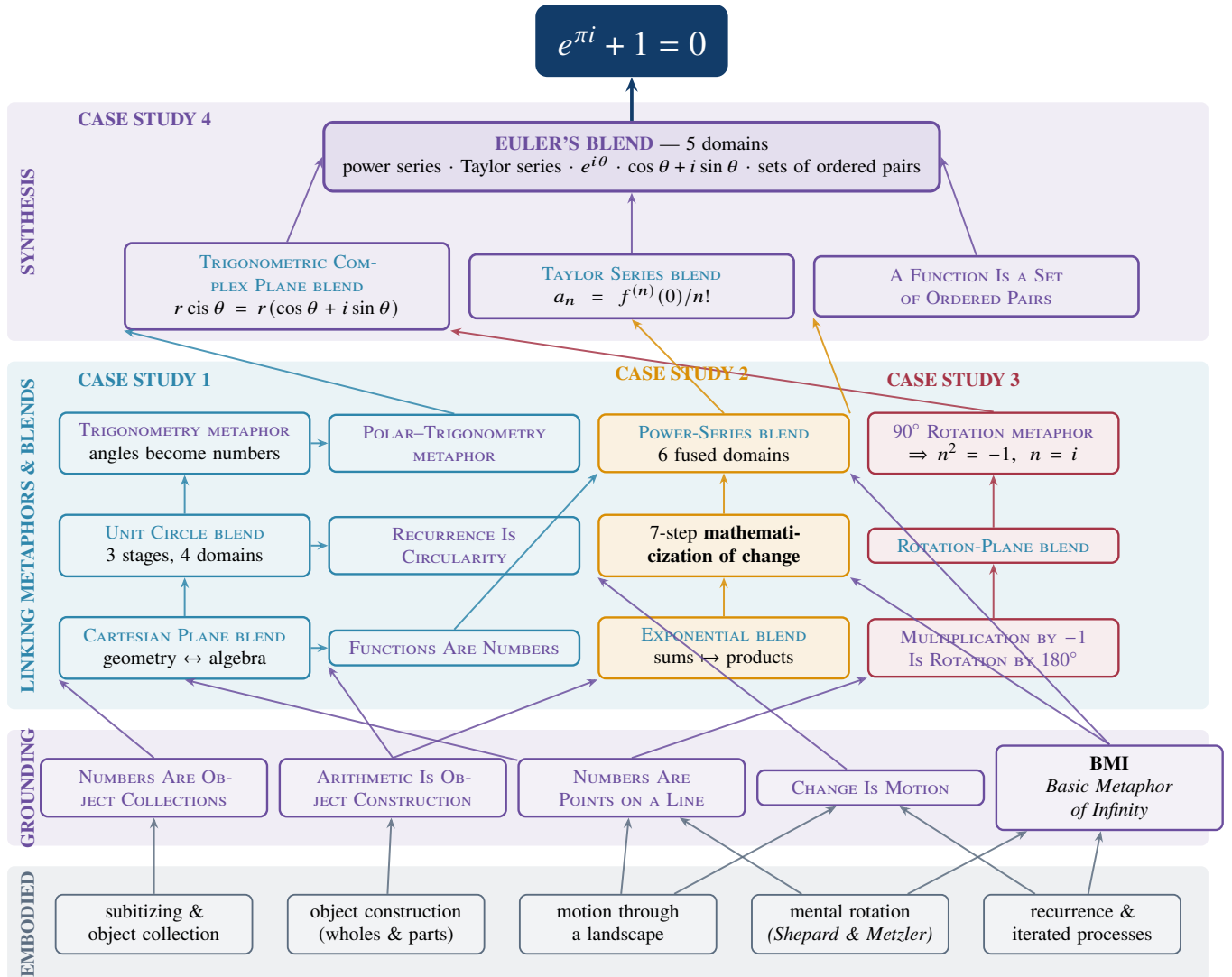


Figure 1. The dependency graph of Part VI. Nothing in the upper bands is a definition adopted for convenience: every one of them is presented by Lakoff & Núñez as *inherited* from the band beneath it. The claim the diagram encodes — and the claim I audit in §6 — is that if you delete the bottom two bands, the top band stops meaning anything.

2 Case Study 1 — Analytic Geometry and Trigonometry

BOOK pp. 383–398

PDF pp. 402–417

DELIVERS: $\cos \pi = -1$ and a second meaning for π

2.1 The Cartesian Plane blend

Descartes' plane is presented not as a definition but as a **blend** of two conceptual domains that share no vocabulary: one has numbers and no points, the other has points and no numbers.

Domain 1 · Number lines	Domain 2 · Euclidean plane, $X \perp Y$
number line x / y	$\leftrightarrow$ line $X /$ line Y
number m on line x	$\leftrightarrow$ line M parallel to Y
the ordered pair (m, n)	$\leftrightarrow$ the point where M intersects N
the ordered pair $(0, 0)$	$\leftrightarrow$ the point where X intersects Y
a function $y = f(x)$	$\leftrightarrow$ a curve
an equation linking x and y	$\leftrightarrow$ a figure
solutions of two simultaneous equations	$\leftrightarrow$ intersection points of two figures
the n -tuple $(n_1, n_2, \dots)$	$\leftrightarrow$ a point in n -dimensional space
the collection of all n -tuples	$\leftrightarrow$ an n -dimensional space

Table 1. The Cartesian Plane blend (book p. 385–386 / pdf p. 404–405). The last two rows are the book's own extension, and they are the cognitive explanation of a familiar teaching phrase: this is *how* we claim to “visualise” a space we cannot see.

Why calling it a blend is not pedantry

A blend has **emergent structure**: entities and entailments present in neither input domain. Nothing in arithmetic contains a point; nothing in Euclid contains a number. Once fused, the composite supports inference in both directions — you reason about curves with algebra and about equations with motion. Lakoff & Núñez's recurring move is to show that what a textbook presents as an *observation* is in fact a *construction* with a describable anatomy.

2.2 FUNCTIONS ARE NUMBERS: the first linking metaphor

Literally, functions are not numbers — addition tables do not contain them. The metaphor maps any arithmetic operation on the y -values of two functions onto the corresponding operation on the *functions themselves*. Their sharpest observation follows immediately:

The two “+” signs are not the same sign (book p. 386 / pdf p. 405)

$$\underbrace{(f + g)}_{\text{metaphorical + created by this mapping}} (x) = \underbrace{f(x) + g(x)}_{\text{literal + addition of two numbers}}$$

This metaphor is load-bearing later: it is what licenses treating a **power series as an infinite sum of functions** in Case Study 2.

2.3 The Unit Circle blend, built in three stages

This is the most carefully staged construction in Part VI, and the staging is the point: at each stage *new entities emerge* that were in neither input.

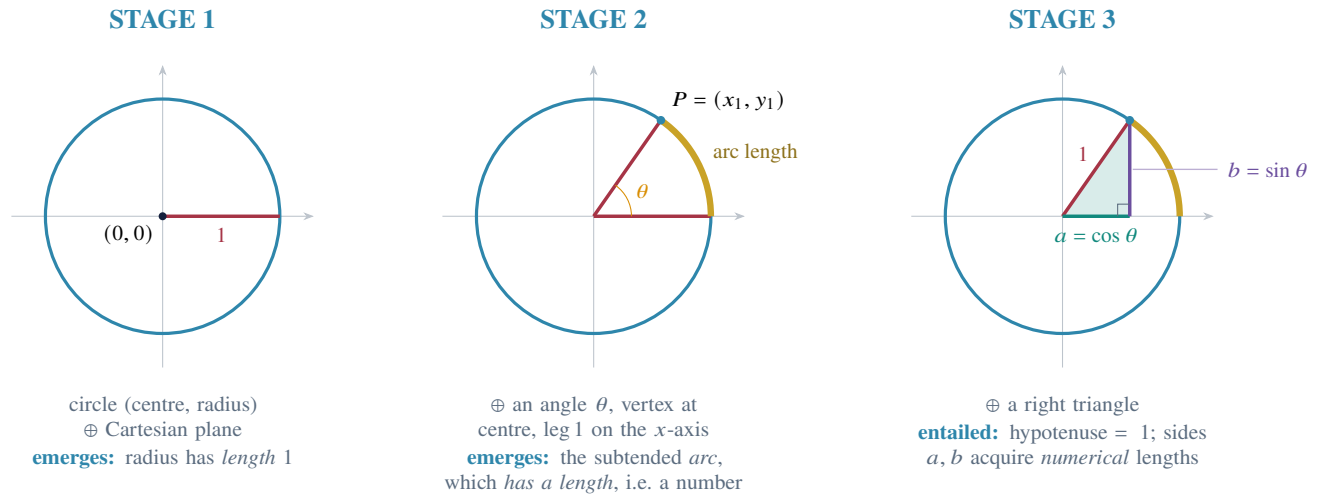


Figure 2. The three stages of the Unit Circle blend, reconstructed from the book's Figures CS 1.2–1.4 and its four-column blend tables (book p. 388–392 / pdf p. 407–411). The final blend fuses **four** domains: circle-with-centre-and-radius, Cartesian plane, angle-with-two-legs, right-triangle-with-hypotenuse.

2.4 The Trigonometry metaphor

Why a metaphor is needed at all

Angles are *not* numbers. You can “add” two angles intuitively by putting their vertices together with a leg in common (the book's Figure CS 1.1) — but that gives you no arithmetic: no subtraction, no division, no raising an angle to a power.

“*In plane geometry, there are angles but no numbers. If we are going to do arithmetic calculation on angles, we have to get angles to be numbers. That is a job for a metaphor.*”

BOOK P. 387 / PDF P. 406

Given the blend, the metaphor itself is startlingly small — three lines:

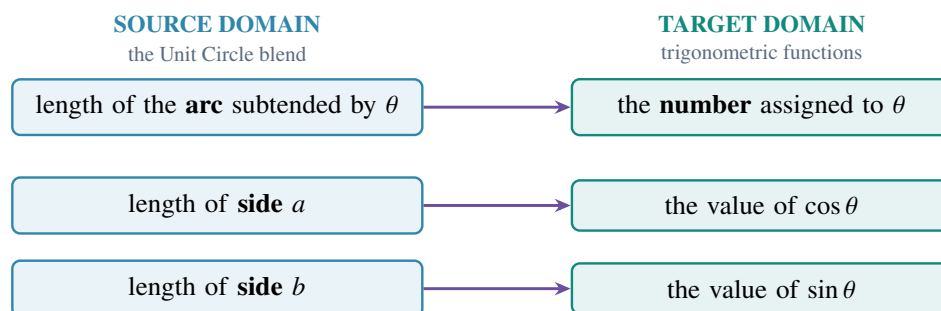


Figure 3. The Trigonometry metaphor (book p. 393 / pdf p. 412). Angles become numbers by being identified with arc lengths — and lengths are already numbers, because the Number-Line blend is inside the Cartesian plane, which is inside the Unit Circle blend.

2.4.1 The entailments — these are consequences, not stipulations

In the blend	Entailed in arithmetic	Source
the unit circle $x^2 + y^2 = 1$	$\Rightarrow \sin^2 \theta + \cos^2 \theta = 1$	Pythagoras, hyp. = 1
θ is a right angle	$\Rightarrow \theta = \pi/2$	circumference = 2π
θ spans half the circle	$\Rightarrow \theta = \pi$	circumference = 2π
θ spans the whole circle	$\Rightarrow \theta = 2\pi$	circumference = 2π
θ is zero	$\Rightarrow \sin 0 = \sin \pi = 0$	blend geometry
θ is two right angles (180°)	$\Rightarrow \cos \pi = -1$	the Euler link

Table 2. Entailments of the Trigonometry metaphor (book p. 393 / pdf p. 412), highlighted row added by me.

The payoff you must not skim past

π 's appearance in Euler's equation is **not** an accident of analysis. It is the arc-length-to-number assignment of this metaphor. And -1 is $\cos \pi$. The book says so explicitly, forty pages before it needs the fact:

“In Case Study 4, where we take up Euler's famous equation $e^{i\pi} + 1 = 0$, we will see that the relationship among π , 0, and 1 in that equation is exactly the relationship given by the Trigonometry metaphor. The relationship between π and -1 in Euler's equation arises from the fact that in the blend, $\cos \pi = -1$.

BOOK P. 394 / PDF P. 413

2.5 RECURRENCE IS CIRCULARITY

Periodicity falls straight out of the blend. The tracing point may go round k times; legs a and b are unchanged; therefore $\sin(\theta + 2k\pi) = \sin \theta$. Unrolled onto the Cartesian plane, that is the wave.

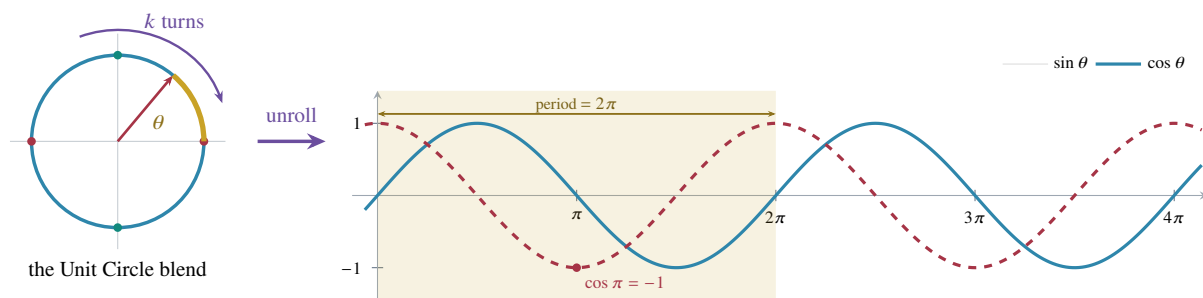


Figure 4. Circularity unrolled into recurrence (the book's Figure CS 1.5, (book p. 395 / pdf p. 414)). The values are confined to the strip $[-1, 1]$ because the radius in the blend is 1 — another entailment rather than a stipulation.

Their further claim is the anthropological one, and it is the moment the book stops being about mathematics only:

“In English and many other languages, the Recurrence Is Circularity metaphor occurs in the conceptual system of the language in general, completely outside mathematics... We speak of recurrent events as happening “over and over”. . . Cultures in which history is believed to recur are spoken of as having “circular concepts of time.”

BOOK P. 395 / PDF P. 414

So the mathematics of recurrence is written in circular functions because human beings already conceptualise recurrence as circularity — not the other way round.

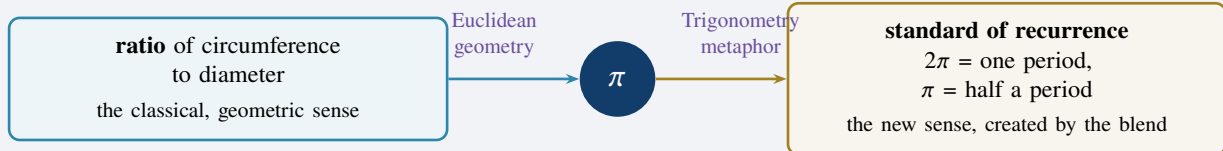
2.6 Polar coordinates, and the second meaning of π

The **POLAR–TRIGONOMETRY METAPHOR** maps $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$. Lakoff & Núñez insist on its metaphoricity in one sentence that is easy to read past and worth underlining:

“Recall that the Cartesian plane in itself has no trigonometry—no angles-as-numbers, no sines and cosines.

BOOK P. 396 / PDF P. 415

The result of Case Study 1: π acquires a second meaning



“Where π was previously only the ratio of the circumference of a circle to its diameter, 2π now becomes a measure of periodicity for recurrent phenomena, with π as the measure of half a period. This is a new idea: a new meaning for π .

BOOK P. 397 / PDF P. 416

Same number. Different idea. Linked by metaphor. And the generalisation they draw — “the meaning of a symbol like π is not merely its numerical value” (book p. 397/pdf p. 416) — is the thesis of the whole Part in miniature.

3 Case Study 2 — What is e ?

BOOK pp. 399–419

PDF pp. 418–438

DELIVERS: e as a *concept*, and why it is 2.71828...

3.1 Demolition: exponentiation is not self-multiplication

The confusion that survives into university teaching

$2^{2/3}$ is **not** 2 multiplied by itself two-thirds of a time. 2^π is **not** 2 multiplied by itself 3.14159... times.

“The exponential mapping for a base b is defined as specified above, but it is not self-multiplication.

BOOK P. 404 / PDF P. 423

Self-multiplication is a *special case that happens to coincide on the integers*, and it rests on a *different* metaphor — **ARITHMETIC IS OBJECT CONSTRUCTION** (a whole made of equal parts). Hence the symbol “ 2^5 ” is conceptually **ambiguous**.

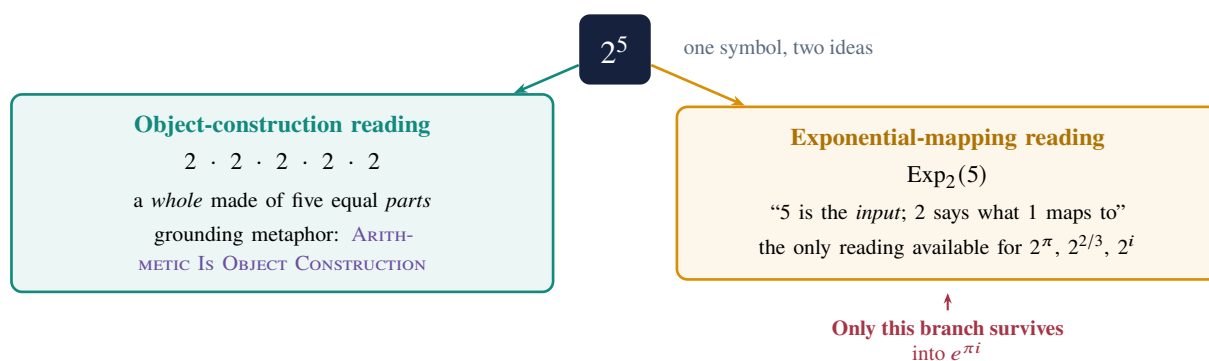


Figure 5. The ambiguity of exponential notation (book p. 404/pdf p. 423). The book proposes writing $\text{Exp}_b(x)$ precisely because the ordinary notation *hides* the asymmetry between b and x .

Where b and x actually live — and why $b^0 = 1$

In “ b^x ” the two symbols are not on the same footing at all:

$\underbrace{b}_{\text{in the range}} \quad \underbrace{x}_{\text{in the domain}} \quad \text{—} \quad b \text{ is merely the number that 1 maps onto.}$

So $b^0 = 1$ is not a convention and not “ b multiplied by itself zero times” (which is nonsense). It says: **the exponential sends the additive identity 0 to the multiplicative identity 1** — true for every base, which is why $b^0 = 1$ always.

“The way that the exponential function is written—“ b^x ”—is misleading. It is done that way for the purpose of ease of calculation, for those cases where self-multiplication will happen to allow you to calculate the value of the function.

BOOK P. 405 / PDF P. 424

3.2 What a logarithm really is

Not “the inverse of a power”, but a **mapping that lets you multiply by adding** — in modern terms a group homomorphism $(\mathbb{R}^+, \times) \rightarrow (\mathbb{R}, +)$. Napier’s labour was tabulating it; the slide rule is the physical instrument of the same idea.

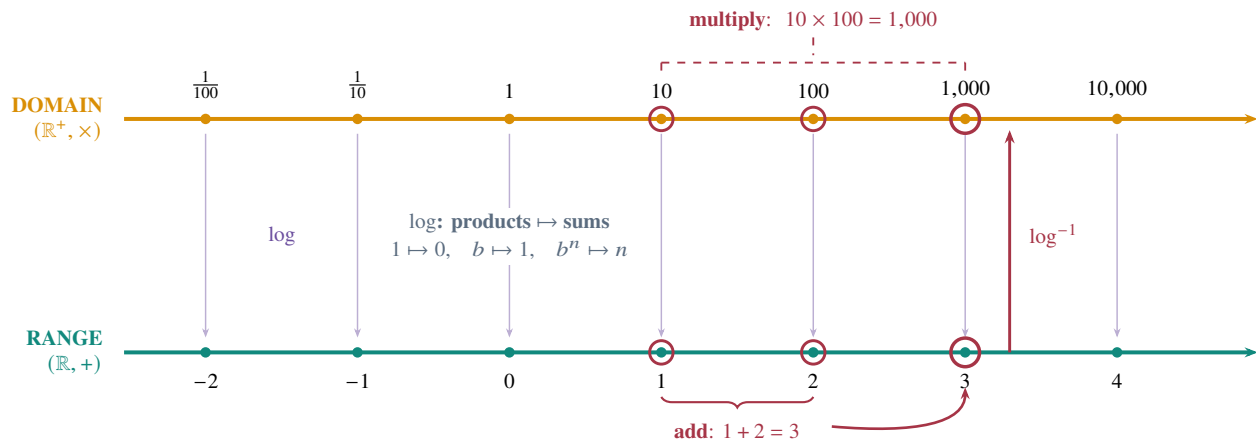
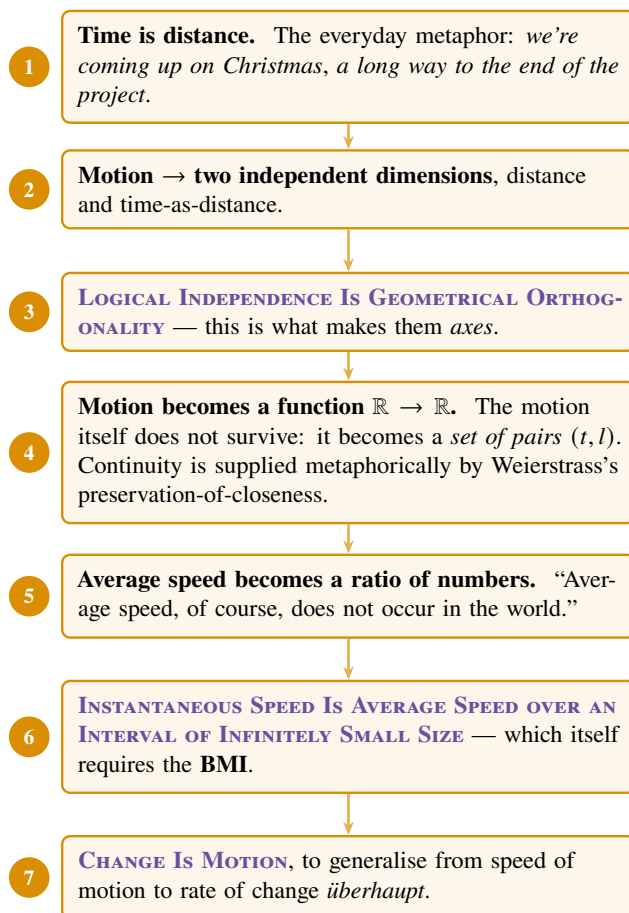


Figure 6. The logarithm mapping, base 10 (book p. 400–402 / pdf p. 419–421) (the book’s Figures CS 2.1–2.2). The four constraints listed in the middle **completely and uniquely determine every value** of the mapping — though, as the book notes, they do not by themselves give an algorithm for computing it.

The exponential is simply this mapping run backwards: $0 \mapsto 1$, $1 \mapsto b$, $n \mapsto b^n$, **sums** $\mapsto$ **products**. Cognitively, Lakoff & Núñez read it as the arithmetization of the metaphor **MULTIPLICATION IS ADDITION** (book p. 405 / pdf p. 424).

3.3 The seven-step mathematicization of change

For a foundations student this is the single most valuable page in Part VI. Before differential calculus can be “the mathematics of change”, **seven** conceptual moves must already have been made — and by the time you meet them in a calculus course they have all been internalised invisibly.



Read the stack downward.

Each step is a *substitution*: something directly perceived is replaced by something calculable. Motion is perceived directly, by motion detectors in the visual system. Nothing in the final object — a set of ordered pairs of real numbers — moves.

What happens if you forget they are metaphors:

“

Motion consists merely in the occupation of different places at different times, subject to continuity... There is no transition from place to place, no consecutive moment or consecutive position, no such thing as velocity except in the sense of a real number, which is the limit of a certain set of quotients.

RUSSELL, *PRINCIPLES OF MATHEMATICS* 1903, p. 473; QUOTED BOOK p. 408 / PDF p. 427

Lakoff & Núñez call this “**pathological literalness**” — mistaking the output of the metaphor chain for the thing itself. Their diagnosis:

“His view that thought was mathematical logic blinded him to the metaphorical mechanisms of mind needed to go from the normal human concept of change to a conceptualization of change in Weierstrass's mathematics.”

BOOK p. 408 / PDF p. 427

Figure 7. The seven steps (book p. 407–408 / pdf p. 426–427). Keep this beside any rigorous treatment of the derivative: it is an inventory of everything you silently import when you say “calculus is the mathematics of change”.

3.4 The derivation — and the exact location of the hinge

Now the payoff. Differentiate a general exponential and watch where the whole of Part VI turns.

Differentiating b^x — the load-bearing step is coloured

$$\begin{aligned}
 \frac{d}{dx}b^x &= \lim_{\Delta x \rightarrow 0} \frac{b^{x+\Delta x} - b^x}{\Delta x} && \text{derivative metaphor + BMI} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{\mathbf{b^x \cdot b^{\Delta x}} - b^x}{\Delta x} && \text{THE HINGE: available \textit{only} because} \\
 &&& \text{the exponential maps sums } \mapsto \text{products} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{b^x (b^{\Delta x} - 1)}{\Delta x} && \text{group the common factor} \\
 &= b^x \cdot \underbrace{\lim_{\Delta x \rightarrow 0} \frac{b^{\Delta x} - 1}{\Delta x}}_{\text{a constant, call it } C} && b^x \text{ contains no } \Delta x, \text{ so it leaves the limit} \\
 &= C b^x
 \end{aligned}$$

The intellectual centre of gravity of the whole Part



A function that maps sums onto products *necessarily* changes in proportion to its own size, and conversely. The two ideas meet at a single line of algebra. Everything in Case Study 4 — including why $\cos y + i \sin y$ turns out to be an exponential at all — runs through this junction.

The concept of e , stated before any number is computed

*“ e ” is the base of the exponential function for which $C = 1$:
 the function whose **rate of change is identical to itself**.*

BOOK P. 410–411 / PDF P. 429–430

Equivalently, unpacked into the book’s own summary box (book p. 415 / pdf p. 434): e^x is the function that (i) maps sums onto products, (ii) maps 1 onto 2.718281828459045 . . . , (iii) is its own derivative, (iv) changes in exact proportion to its size.

3.5 Why 2.718281828459045...? The Power-Series blend

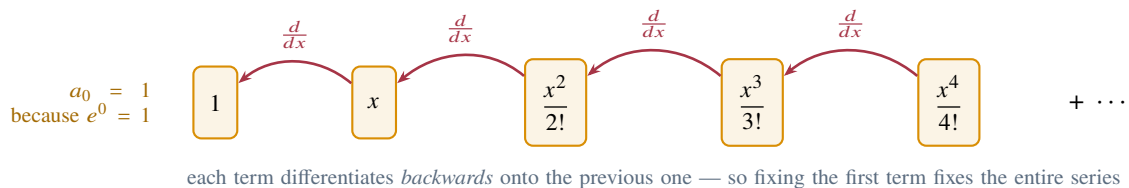
Euler's own route to the number goes through power series, and the book's job is to make explicit the blend that is taken for granted whenever a textbook differentiates a series term by term. It fuses **six** domains.

1 Wholes & parts	2 Numbers	3 Functions	4 Infinite sums via BMI	5 Power series via BMI	6 Calculus applied
Whole	resulting number c	composite function $h(x)$	infinite sum $S = \sum_{k=0}^{\infty} a_k$	function at infinity $F(x) = \sum a_k x^k$	derivative at infinity $F'(x) = \sum k a_k x^{k-1}$
Parts	operand numbers a, b	component functions $f(x), g(x)$	finite terms a_k	finite terms $a_k x^k$	derivatives of finite terms $k a_k x^{k-1}$
part-whole relation	$a + b = c$	$f(x) + g(x) = h(x)$	$a_{k-1} + a_k$	$a_{k-1} x^{k-1} + a_k x^k$	$(k-1)a_{k-1} x^{k-2} + k a_k x^{k-1}$
—	partial sum	partial sum of functions	$S_n = \sum_{k=0}^n a_k$	$F_n(x) = \sum_{k=0}^n a_k x^k$	$F'_n(x) = \sum_{k=0}^n k a_k x^{k-1}$

Table 3. The Power-Series blend (book p. 413/pdf p. 432). Column 1→2 is **ARITHMETIC IS OBJECT CONSTRUCTION**; 2→3 is **FUNCTIONS ARE NUMBERS** (from Case Study 1); 2→4 is the **BMI**; 4→5 is **FUNCTIONS ARE NUMBERS** again, now applied to infinite sums; 6 applies differentiation to an object that only exists because of columns 1–5.

3.5.1 The series is forced, not guessed

Starting from “ e^x is its own derivative” plus $\frac{d}{dx} a_n x^n = n a_n x^{n-1}$, each term must be the derivative of the term after it. And $x = 0$ kills every term but the first, while $e^0 = 1$, so $a_0 = 1$. Everything else cascades:



$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad \Rightarrow \quad e = e^1 = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots = 2.718281828459045 \dots$$

$$\sum_{k=0}^{n-1} 1/k! \rightarrow e$$

the textbook limit, converging *much* more slowly

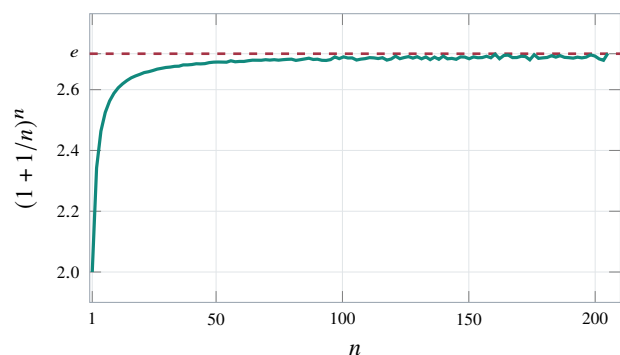
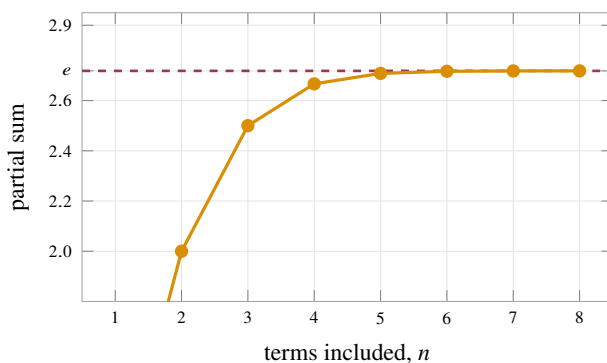


Figure 8. Two routes to the same number (book p. 414–419/pdf p. 433–438). The contrast is itself an argument of the book's: the power series comes *out of the concept* (“ e^x is its own derivative”), whereas $(1 + 1/n)^n$ is normally handed to students as a definition with no explanation. Note also how much faster the factorial series converges.

3.6 The appendix: the best-argued pages in Part VI

The question textbooks never ask (book p. 416 / pdf p. 435)

“Why should this particular limit, $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$, be the base of the exponential function that is its own derivative?

BOOK P. 416 / PDF P. 435

Their answer is a five-link chain, and it starts from the *meaning* of e , not from a definition:

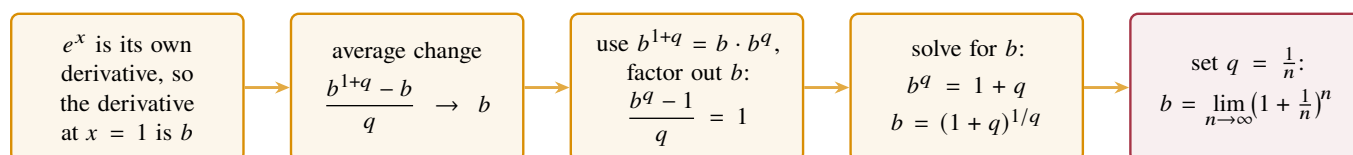


Figure 9. The derivation of the familiar limit *from* the concept of e (book p. 416–419 / pdf p. 435–438). The book’s own summary of what the result rests on: “(1) what e means, (2) the metaphors for mathematicizing change, (3) the limit metaphor, which is a special case of the BMI, (4) some simple algebraic manipulations.”

Their moral, and it is a good one

“A system of conceptual metaphors and blends characterizes systematic relationships between concepts and formulas. We have to understand the nature of that system if we are to understand the mathematical ideas embodied by the formulas.

BOOK P. 419 / PDF P. 438

4 Case Study 3 — What is i ?

BOOK pp. 420–432

PDF pp. 439–451

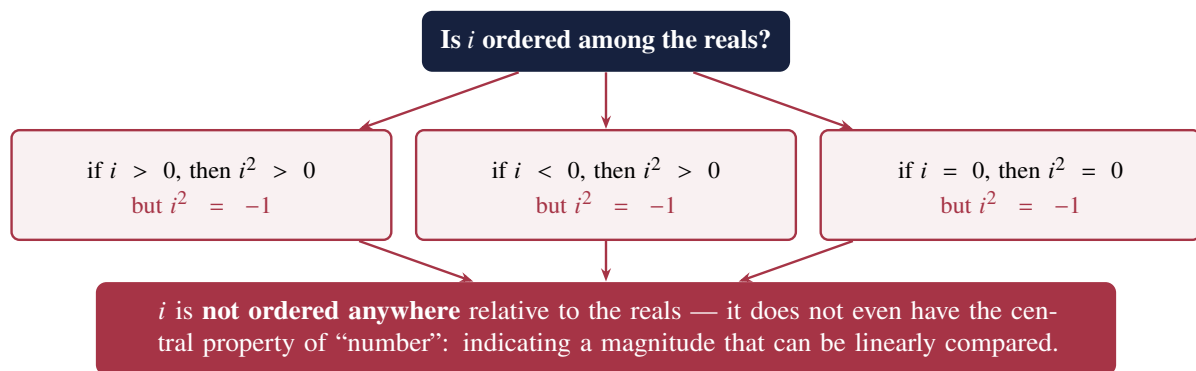
DELIVERS: the complex plane *derived*, not stipulated

4.1 A cognitive diagnosis of three centuries of resistance

“Mathematicians simply could not make it fit their idea of what a number was supposed to be. A number was supposed to be a magnitude. $\sqrt{-1}$ is not a magnitude comparable to the magnitudes of real numbers. No tree can be $\sqrt{-1}$ units high. You cannot owe someone $\sqrt{-1}$ dollars.

BOOK P. 420 / PDF P. 439

The book then runs the elementary trichotomy argument — and the point is not the argument (which is trivial) but *what it costs*:



What i loses — measured against the grounding metaphors

Property inherited from...	N	Z, Q	R	i
subitizable (perceived at a glance)	✓	×	×	×
a collection of physical objects	✓	×	×	×
a magnitude, linearly ordered	✓	✓	✓	×
obeys closure & the laws of arithmetic	✓	✓	✓	✓

Only the last row survives. As the book puts it (book p. 422/pdf p. 441): “If i is to be a number, it is a number only by virtue of closure and the laws of arithmetic.”

4.2 Closure as the engine of every extension

Their historical-cognitive claim is worth stating carefully, because it is the load-bearing thesis of this case study. Closure is an *entailment* of the four grounding metaphors for arithmetic — it arises naturally when only $+$ and $\times$ are in view. Greek-style essentialism then *reanalyses* it as part of the **essence** of “number”. After that, every extension is forced.

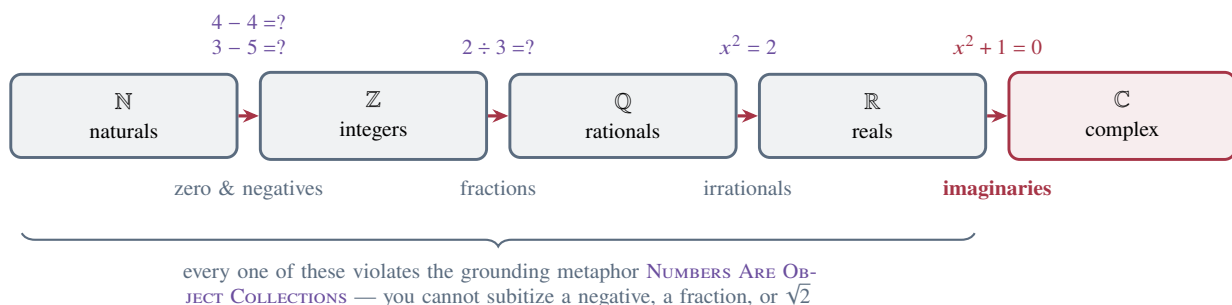
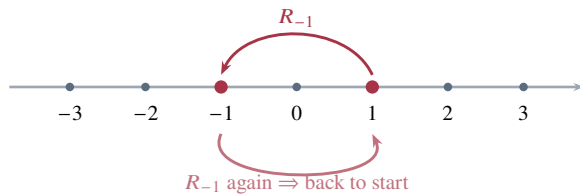


Figure 10. The closure ladder (book p. 421–423 / pdf p. 440–442). The book’s summary of the tension: “the technical idea that closure is part of the essence of arithmetic operations on ‘numbers’ has been in conflict with ‘number’ as characterized by subitizing and the grounding metaphors. But within technical mathematics, the imperative of essence implied the imperative of closure.”

4.3 The construction of the complex plane from rotation

This is the most elegant piece of reverse-engineering in the book: the complex plane is *derived* in four moves from a fact about mental rotation.

MOVE 1. The flip on the number line



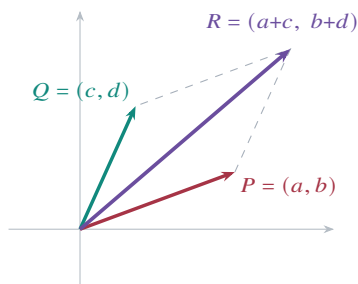
$-n$ and $+n$ are *symmetric about the origin*. Cognitively we register that symmetry as a **mental rotation** — flipping the positive half-line onto the negative half-line, preserving distance. Applying it twice returns you to where you started: $R_{-1}(R_{-1}(n)) = n$.

MULTIPLICATION BY -1 IS ROTATION TO THE SYMMETRY POINT

“The spatial operation R_{-1} is part of the cognitive apparatus we use to conceptualize the relationship between positive and negative numbers. It is not part of the formal mathematics.”

BOOK P. 425 / PDF P. 444

MOVE 2. Addition falls out of Euclid



The **ROTATION-PLANE BLEND** contains the Euclidean plane. So a theorem of the Euclidean plane is inherited free of charge:

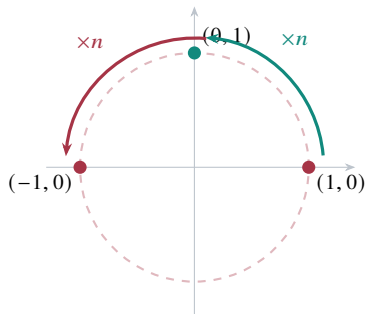
“Each pair of intersecting line segments uniquely determines a parallelogram.”

Place O at the origin and let the segments be OP and OQ . The fourth vertex R is then *uniquely determined*, and its coordinates are the coordinate-wise sums.

Vector addition of complex numbers is not a stipulation — it is inherited geometry. BOOK P. 426–427 / PDF P. 445–446, FIG. CS 3.1

Figure 11. Moves 1 and 2 of the derivation. The **ROTATION-PLANE BLEND** combines the **ROTATION-NUMBER-LINE BLEND** with the **CARTESIAN PLANE BLEND**; in it, rotation by 180° is multiplication by -1 , and it applies to *every* line through the origin, not just the x -axis.

MOVE 3. Give 90° an arithmetic correlate



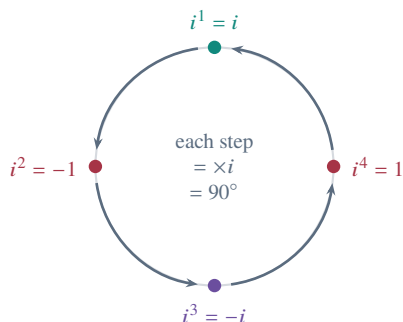
In the rotation plane, rotation by 180° is multiplication by a number. Rotation by 90° has **no arithmetic correlate at all**. So supply one — that is the whole of the new metaphor:

SPACE		ARITHMETIC
rotation by 90°	$\mapsto$	multiplication by n
two rotations by 90°	$\mapsto$	multiplication by n^2

Now compute. 90° applied to $(1, 0)$ gives $(0, 1)$; a second 90° gives $(-1, 0)$. So $n^2 \cdot (1, 0) = (-1, 0) \cdot (1, 0)$, hence

$$n^2 = -1, \quad n = \sqrt{-1} = i$$

MOVE 4. The four-cycle



The multiplication rule, derived in four lines

$$\begin{aligned}
 (a + bi)(c + di) &= (a + bi)c + (a + bi)di && \text{distributive law} \\
 &= ac + bci + adi + bd i^2 && \text{distributive law} \\
 &= ac + bci + adi - bd && \text{by } i^2 = -1 \\
 &= (ac - bd) + (bc + ad)i && \text{associative law}
 \end{aligned}$$

Nothing here is postulated. Distributivity, associativity and commutativity are already present because we are inside a special case of the **CARTESIAN PLANE BLEND**; $i^2 = -1$ came out of Move 3. (book p. 430 / pdf p. 449)

Why the i -axis is at 90° — the question formal presentations cannot answer

Multiplication by -1 is rotation by 180° . A rotation of 180° is the result of two 90° rotations. Since $i \cdot i = -1$, **multiplication by i must be rotation by 90°** — two of them give 180° , i.e. multiplication by -1 .

“*The complex plane is just the 90° rotation plane—the rotation plane with the structure imposed by the 90° Rotation metaphor added to it. Multiplication by i is “just” rotation by 90° . This is not arbitrary; it makes sense.*

BOOK P. 429 / PDF P. 448

4.4 Three ways to present $\mathbb{C}$ — and what each one costs

The book closes Case Study 3 by comparing its own account with the two that are actually taught. This table is mine, built from their discussion (book p. 431–432/pdf p. 450–451).

	A. Rotation account (the book's)	B. Plane + rules	C. Pure ordered pairs (Dedekind–Weierstrass spirit)
Why is the i -axis at 90° ?	Derived from $i^2 = -1$ and $\times(-1) = 180^\circ$	Unexplained — it is drawn that way	No axes exist
Why does addition work like vectors?	Euclid's parallelogram theorem , inherited	Stipulated rule	Stipulated rule
Why that multiplication rule?	Four lines of algebra from $i^2 = -1$	Stipulated: $(ac-bd, bc+ad)$	Stipulated
Geometry present?	yes, constitutively	yes, decoratively	none at all
What it is good for	understanding	calculation	proof , generalisation, rigour

Table 4. My comparison, from the book's closing argument (book p. 431–432/pdf p. 450–451). Their verdict on B and C: “they give very little insight into the conceptual structure of the complex numbers—into *why* they are the way they are.” The rightmost column is not a criticism of formal mathematics; it is a statement about a division of labour.

The key word is *motivate* — and the crux is proof vs. understanding

“*The crux of the issue here is the difference between proof and understanding. As our quotation from Benjamin Peirce (in Case Study 1) indicates, it is possible to prove a deep mathematical result without understanding it at all. . . For us, mathematics is not just about proof and computation. It is about ideas.*

BOOK P. 431 / PDF P. 450

They anticipate the obvious objection — *aren't rotations just a way of visualising $\sqrt{-1}$, useful for pedagogy but not part of the mathematics?* — and answer no, on the ground that the rotation account is *true given the conceptual metaphors and blends commonplace in classical mathematics*, and that it alone reveals the structure.

5 Case Study 4 — $e^{\pi i} + 1 = 0$

BOOK pp. 433–452

PDF pp. 452–471

DELIVERS: why the equation is true by virtue of what it means

5.1 Dissolving the pseudo-question

“What does it mean to multiply e by itself πi times?”

Nothing. The question is malformed.

“ $e^{\pi i}$ is not an instance of the self-multiplication function. It is, rather, a special case of the exponential function $w = e^z$, where $z = \pi i$.

BOOK P. 434 / PDF P. 453

πi is the point $(0, \pi)$ in an input complex plane; the function $w = e^z$ sends it to the point $(-1, 0)$ in an output complex plane.

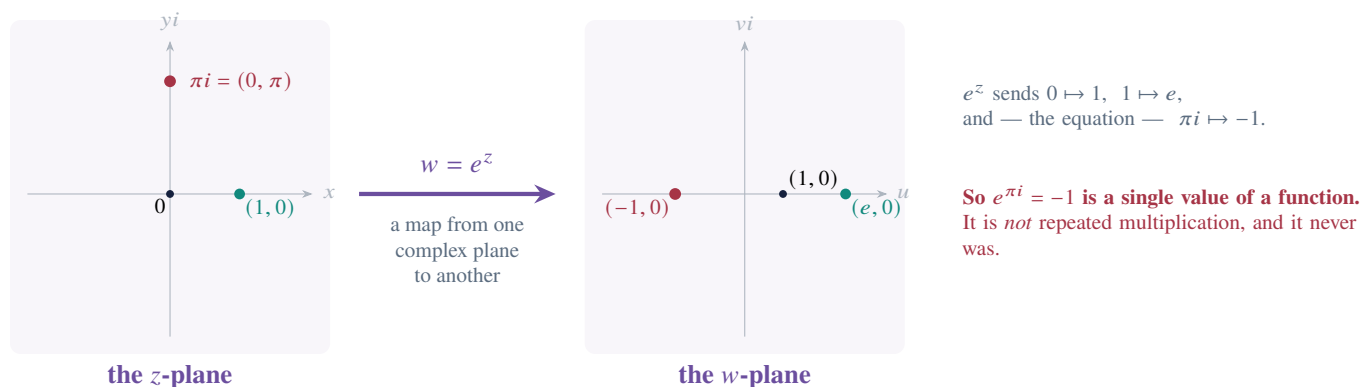


Figure 12. The book’s Figure CS 4.1 (book p. 434–435 / pdf p. 453–454).

5.2 Their own anti-climax — and why it matters

To their real credit, Lakoff & Núñez immediately refuse to let the reader be impressed by this.

“This ought to leave you cold... Numerically, all that is happening is that $3.141592654 \dots$ on the y -axis in one plane is being mapped by some function onto -1 on the x -axis in another. . . From a purely numerical point of view, this has no interest at all! If all that is at issue is calculating values of functions and seeing which number maps onto which other number, the greatest equation of all time is a big bore! Euler was too smart to care about something that boring.

BOOK P. 434 / PDF P. 453

The move that organises the rest of the case study

The significance of $e^{\pi i} = -1$ is **conceptual, not numerical**. It is not given by the values computed for $w = e^z$. It lies in what the equation tells us about how the branches of mathematics are related — algebra to geometry, geometry to trigonometry, calculus to trigonometry, and complex arithmetic to all of them (book p. 434 / pdf p. 453).

5.3 The Trigonometric Complex Plane blend

Case Study 1's **POLAR-TRIGONOMETRY BLEND** $\oplus$ Case Study 3's **90° ROTATION-PLANE BLEND**. The entailments are the trigonometric form of complex arithmetic:

Entailments in the blend (*book p. 437 / pdf p. 456*)

$$r \cos \theta = a \quad r(i \sin \theta) = bi \quad \underbrace{r \operatorname{cis} \theta}_{\text{notation}} : r(\cos \theta + i \sin \theta) = a + bi$$

$$(r_1 \operatorname{cis} \alpha) \cdot (r_2 \operatorname{cis} \beta) = (r_1 r_2) \operatorname{cis}(\alpha + \beta) \iff (a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Do not let the familiarity hide the metaphor

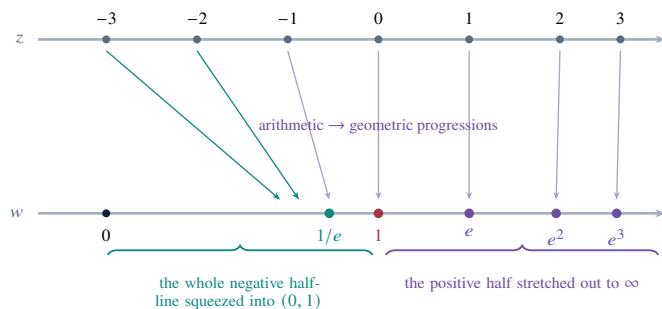
“*The complex plane, with complex addition and multiplication, has no inherent trigonometric structure. Complex arithmetic does not need trigonometry. Trigonometry is in another mathematical domain altogether. When we understand complex arithmetic in terms of trigonometry, we are conceptualizing one mathematical domain in terms of another, which is what conceptual metaphor does.*

BOOK P. 438 / PDF P. 457

5.4 Two pieces of the function e^z

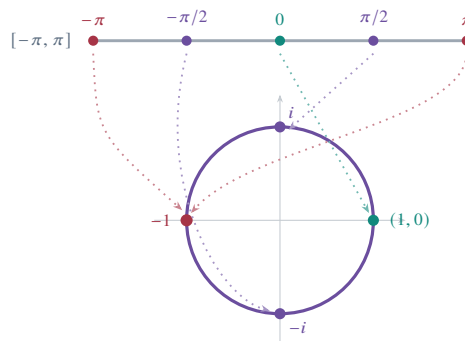
Since $e^z = e^{x+yi} = e^x \cdot e^{yi}$, the function splits into two halves that can be understood separately and then recombined.

(a) The $y = 0$ slice: e^x on the real axis



Label each image point with its pre-image and you have literally drawn a **slide rule** — the logarithm of Case Study 2 reappearing as a picture (book p. 441 / pdf p. 460), the book's Figure CS 4.5.

(b) The wrapping function $f(\theta) = \cos \theta + i \sin \theta$



The segment is **wrapped around the unit circle**. Both π and $-\pi$ land on $(-1, 0)$ — that *is* periodicity. Points a distance π apart map to points related by $\times(-1)$; points $\pi/2$ apart, by $\times i$. **Case Study 3's rotation metaphor is now doing visible work inside a trigonometric object** (book p. 439–440 / pdf p. 458–459).

Figure 13. The two halves of $e^z = e^x \cdot e^{yi}$.

5.5 Why $e^{iy} = \cos y + i \sin y$

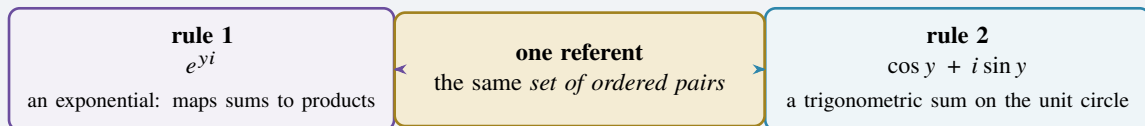
The standard route is via Taylor series, and the book runs it honestly: two conceptually different rules turn out to have the *same derivatives at 0*, cycling with period four.

n	$f(y) = \cos y + i \sin y$		$f(y) = e^{yi}$	
	derivative	at $y = 0$	derivative	at $y = 0$
0	$\cos y + i \sin y$	1	e^{yi}	1
1	$-\sin y + i \cos y$	i	$i e^{yi}$	i
2	$-\cos y - i \sin y$	-1	$-e^{yi}$	-1
3	$\sin y - i \cos y$	$-i$	$-i e^{yi}$	$-i$
4	$\cos y + i \sin y$ (back to start)	1	e^{yi} (back to start)	1

Table 5. The two derivative cycles (book p. 445–446 / pdf p. 464–465). Same column of values $\Rightarrow$ same Taylor series $\Rightarrow$ same set of ordered pairs.

What the “=” in $e^{iy} = \cos y + i \sin y$ actually means

This is the cognitive gloss that generalises far beyond Euler’s formula. Under the metaphor **A FUNCTION IS A SET OF ORDERED PAIRS**, the equals sign does **not** say that the two sides express the same idea.



“*The “=” means that the two conceptually different mapping rules on the two sides of the equal sign have the same referent, the same set of ordered pairs.*

BOOK P. 446 / PDF P. 465

Worth internalising: much of the apparent magic of mathematical identities is the collapse of *same rule* into *same graph*. The book flags this back to Chapter 16, where “=” is shown to relate ideas that are cognitively very different.

Setting $y = \pi$ and using $\cos \pi = -1$ (the Trigonometry metaphor, Case Study 1) and $\sin \pi = 0$:

$$e^{\pi i} = \cos \pi + i \sin \pi = -1 + 0i = -1 \quad \Rightarrow \quad e^{\pi i} + 1 = 0$$

5.6 Euler’s blend, and the answer that was worth waiting for

The Taylor-series story tells you *that* the equation holds. It does not tell you *why it should*. The book’s own verdict on its own exposition is disarming:

“At first, all this sounds like mumbo-jumbo... Okay. But again, why does it matter that $e^{yi} = -1$ when $y = \pi$? And what does it mean?

BOOK P. 446 / PDF P. 465

The answer comes in two parts: a blend, and then — finally — four shared properties.

5.6.1 Euler’s blend: five domains

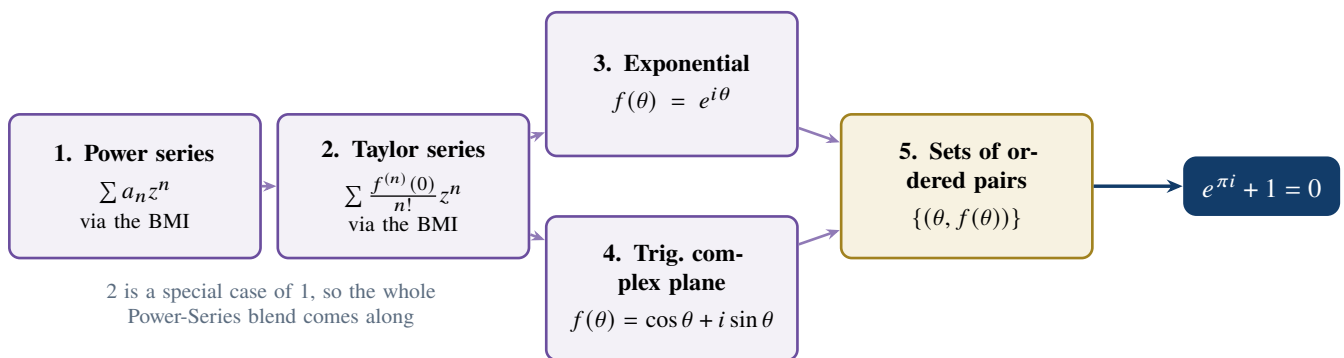


Figure 14. Euler’s blend (book p. 447 / pdf p. 466). The book names it “partly in honor of Euler and partly because we suspect that he must have had, at least intuitively and unconsciously, an implicit understanding of the relationships between the branches of classical mathematics that the blend represents.”

5.6.2 The real answer: four properties, held in common

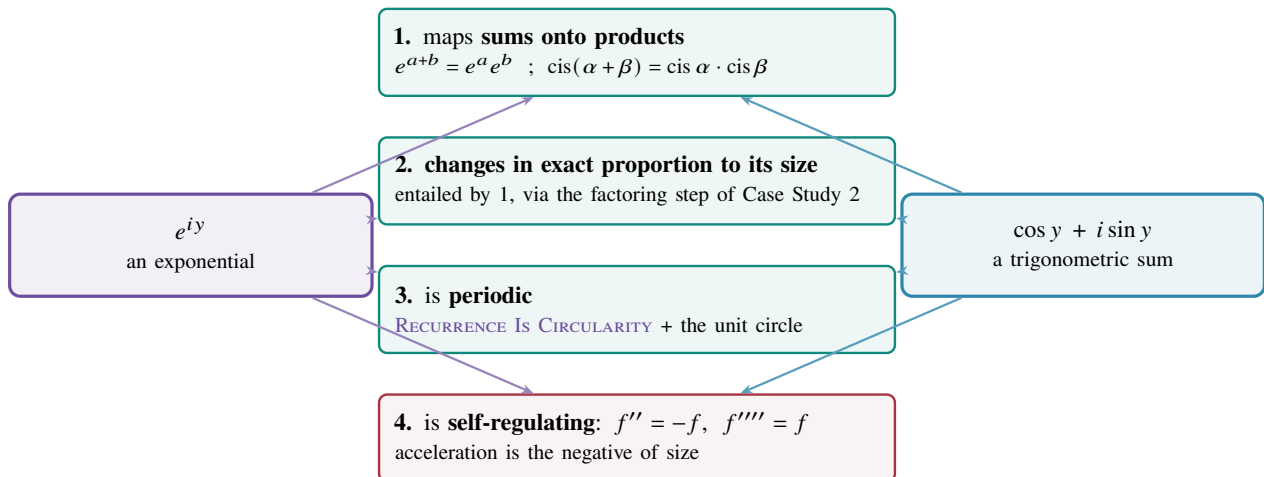


Figure 15. The idea analysis of the identity (book p. 448–450 / pdf p. 467–469). Two functions characterised by *different combinations of concepts entail exactly the same properties*.

The sentence the whole book is built to deliver

“Though they are characterized by different combinations of concepts, they entail, by virtue of what they mean, exactly the same properties. From the perspective of mathematical idea analysis, this is why they are equal.

BOOK P. 450 / PDF P. 469

5.6.3 Self-regulation, made visible

Property 4 is the one that is easiest to state and hardest to *see*. $f'' = -f$ means the acceleration is the negative of the size: as the function grows, its deceleration grows, until growth stops and reverses.

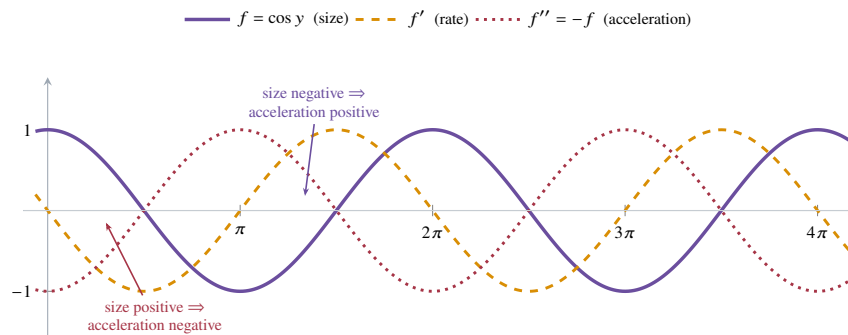


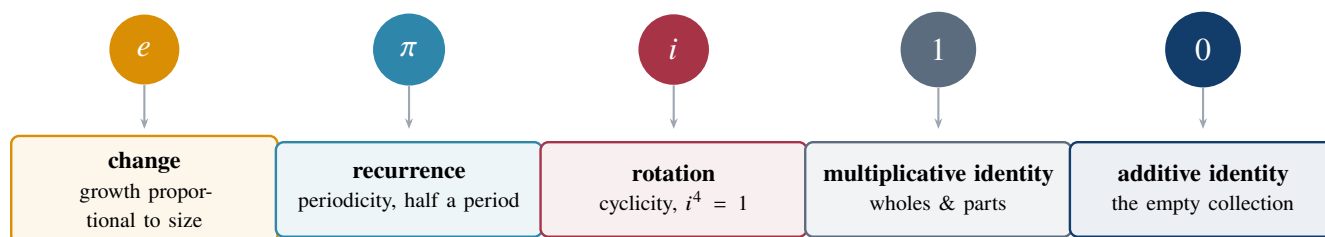
Figure 16. “The self-regulating character comes from the fact that the acceleration is the negative of the size” (book p. 448 / pdf p. 467). This is the harmonic oscillator, and the book’s contribution is to give it a *conceptual* rather than merely formal presence inside $e^{i\theta}$.

5.7 The philosophical punchline

Why e , π , i , 1 and 0 — and not 192,563,947.9853294867

“They are not mere numbers; they are the arithmetizations of concepts... Unlike, say, 192,563,947.9853294867, these numbers have conceptual meanings in a system of common, important nonmathematical concepts, like change, acceleration, recurrence, and self-regulation.

BOOK P. 450–451 / PDF P. 469–470



These concepts matter in everyday life. **That** is why these numbers keep turning up — and why the equation that binds them is worth caring about.

Figure 17. The closing argument of Part VI (book p. 450–451 / pdf p. 469–470).

The last words of Part VI

“It is no accident that our branches of mathematics are linked in the way they are. Those conceptual connections—via conceptual metaphors and blends—express ideas that matter to us. The way the branches of mathematics are interrelated is a consequence of what is important to us in our everyday lives and how we conceptualize those concerns. The various branches of mathematics mathematicize our concerns. The mechanism for doing so is the all too human conceptual metaphor and blending.

BOOK P. 451 / PDF P. 470

6 Assessment: what holds up, and where it leaks

A dossier that only reported the book's claims would be of no use to you. What follows is my own audit. I have separated it into what I take to be **secure**, and **six specific places where the argument does not carry the weight put on it**. Read the second list with a pencil.

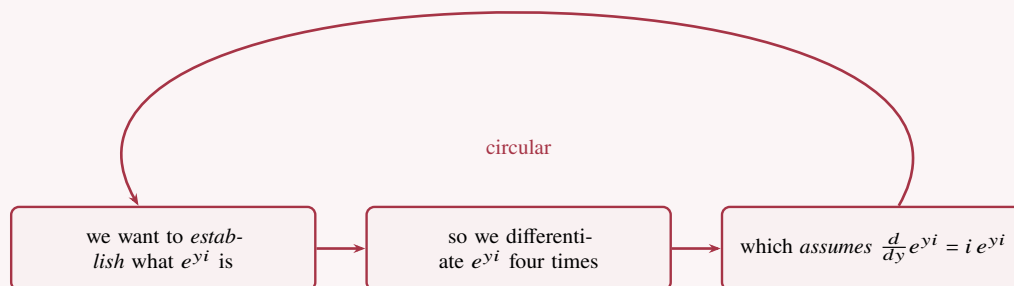
6.1 Secure — and genuinely worth having

Claim	Why it stands
1 Exponentiation $\neq$ self-multiplication; $\text{Exp}_b(x)$ exposes the domain/range asymmetry	A real conceptual confusion, and the reframing makes $b^0 = 1$ stop being a convention. Nothing here depends on the metaphor theory being right.
2 “Maps sums to products” and “rate of change $\propto$ size” are <i>mutually entailing</i> , and the entailment is localised in one algebraic step	This is a genuine mathematical insight served as a conceptual one. It is checkable and it is correct.
3 The rotation derivation of $\mathbb{C}$: addition from Euclid's parallelogram, i from giving 90° an arithmetic correlate	Better motivation than “consider $\mathbb{R}[x]/(x^2 + 1)$ ” — and it predicts, rather than stipulates, the multiplication rule.
4 π has two meanings (ratio; half-period) linked by metaphor, not coincidence	Makes sense of why π appears in Fourier analysis, signal processing and Euler's formula at all.
5 “ $=$ ” in $e^{iy} = \cos y + i \sin y$ asserts <i>co-reference of distinct rules</i> , not identity of ideas	Generalises far beyond Euler. This is the single most transferable idea in Part VI.
6 The self-regulation reading of $f'' = -f$	Gives the harmonic oscillator a conceptual presence inside $e^{i\theta}$ instead of a merely formal one.

6.2 Six places the argument leaks

LEAK 1 A circularity in the Taylor-series demonstration (book p. 446 / pdf p. 465)

To compute the derivatives of $f(y) = e^{yi}$ the book writes $f'(y) = i e^{yi}$, citing the real-case rule $\frac{d}{dy} e^{yc} = c e^{yc}$. **But the complex exponential has not been defined at that point in the text** — defining it is precisely what the Taylor-series argument is supposed to accomplish.



The standard fix: define $e^z := \sum_{n \geq 0} z^n / n!$ and derive everything from there. But then the “idea analysis” is reconstructing a *definition*, not discovering a fact — a weaker and more honest claim than the one the chapter makes.

LEAK 2 Analyticity is waved through

“Under certain conditions for convergence” appears *once*, in passing (book p. 442 / pdf p. 461). But

same derivatives at 0 $\implies$ same Taylor series $\implies$ same function

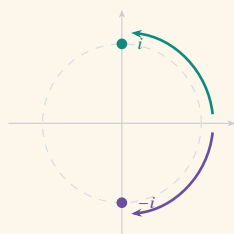
is **false** for smooth-but-not-analytic functions. The standard counterexample:

$$g(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases} \quad g^{(n)}(0) = 0 \text{ for all } n, \text{ yet } g \neq 0.$$

The conclusion *is* true in Euler's case — but the reason it is true is analytic, not metaphorical, and the book does not supply it.

LEAK 3 The 90° metaphor underdetermines i — and this is interesting

Rotate counterclockwise and you get i ; rotate clockwise and you get $-i$. **Both satisfy** $n^2 = -1$. The book does not flag the ambiguity.



But this is **not a flaw in the mathematics** — it is exactly complex conjugation, the non-trivial automorphism of $\mathbb{C}$ over $\mathbb{R}$. I would count it as evidence *for* the program: the metaphor's residual ambiguity tracks a real algebraic symmetry. Worth a margin note in your copy.

LEAK 4 “ i as a metaphorical proportionality factor” is hand-waving

At (book p. 448/pdf p. 467), item 1(c), the book says the derivative of e^{yi} is proportional to the function “with i being the factor expressing that proportionality” and adds, parenthetically, “*Here the ‘proportion’ is metaphorical: It is $\sqrt{-1} = i$. We will discuss the meaning of this metaphorical value below.*” The discussion that follows (self-regulation) is good, but it never explains **in what sense $\sqrt{-1}$ is a proportion**. A proportion is a ratio of magnitudes, and the book has spent Case Study 3 arguing at length that i has no magnitude. The gap is left open.

LEAK 5 Post-hoc reconstruction is presented with the authority of data

The blends arrive as tables, in the format of findings. But **nothing in Part VI is an experiment**. Two very different claims are never separated:

STRONG claim

This is the conceptual structure present in a mathematician's mind.

Would need converging evidence: priming studies, gesture analysis, error patterns, neuroimaging. None is offered in Part VI.

WEAK claim

This is a coherent rational reconstruction that explains why the axioms look as they do.

Fully supported by the text — and still valuable.

The chapters read as if establishing the strong claim while in fact delivering the weak one. **Read Part VI as a philosophy of mathematical understanding, not as cognitive science of mathematicians.** Mathematicians' reviews of the book have pressed on exactly this point, and also on scattered technical slips elsewhere in the volume.

LEAK 6 “Truth is relative to a conceptual structure” is undefended

The claim at (book p. 436/pdf p. 455) admits two readings, and the book does not choose:

Defensible reading

The *meaningfulness* of a sentence is relative to a conceptual structure. $e^{\pi i} + 1 = 0$ says nothing inside Euclidean geometry because the symbols have no referents there.

Uncontroversial, and enough for everything Part VI actually needs.

Indefensible reading

The *truth* of a sentence is relative to a conceptual structure — so the equation could have come out otherwise had we conceptualised differently.

A strong anti-realist thesis, asserted rather than argued.

The rhetoric of the chapter leans on the second; the arguments only ever establish the first.

6.3 Net verdict**Take it as**

- a theory of **mathematical understanding**
- a generator of better motivations for definitions
- a discipline for asking “why *these* axioms?”

Do not take it as

- empirical cognitive science
- a replacement for proof
- a settled philosophy of mathematical truth

Density of original content

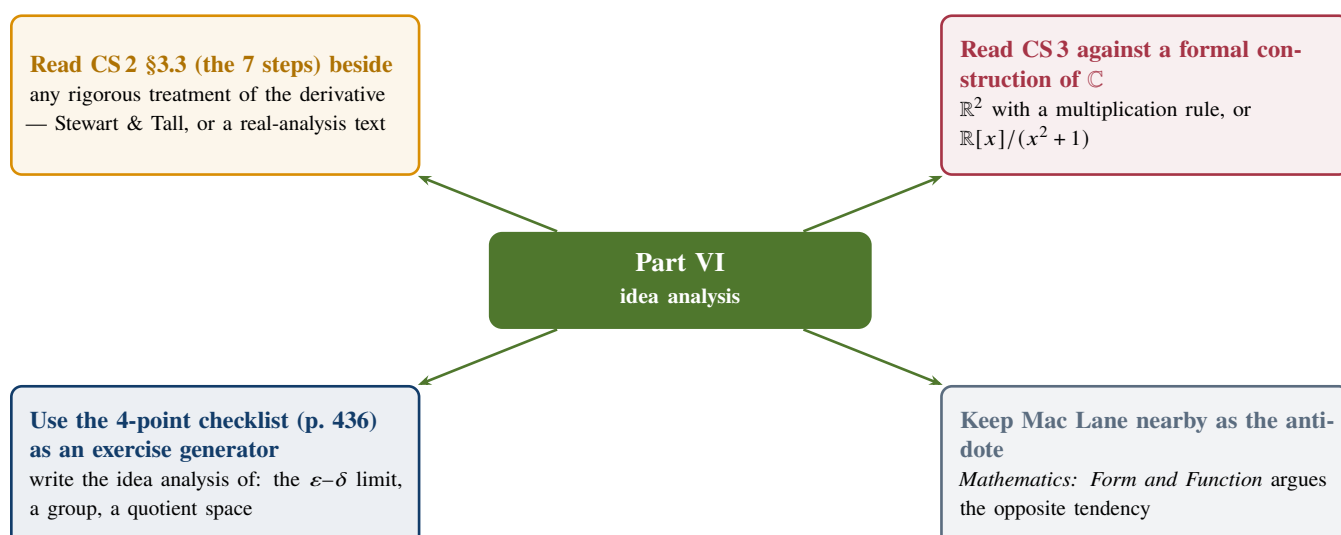
CS 2 and CS 3 carry the most.

CS 1 is setup.

CS 4 mostly re-runs standard analysis; its payoff is § “What Are the Ideas?”, pp. 446–451.

7 How to use this against your own shelf

You are reading Part VI inside a foundations self-study programme, alongside Stewart & Tall, Courant & Robbins, Mac Lane, Devlin, Tao and a shelf of logic texts. Part VI is not a substitute for any of them. It is the **counterweight**: the book that asks *why these axioms and not others*, where the formal texts ask *what follows from these axioms*.



7.1 A concrete four-session plan

Session	Read	Do	Question to answer in writing
1	CS 1, book pp. 383–398	Redraw the three-stage Unit Circle blend from memory	Which properties of sin and cos are <i>entailments</i> of the blend, and which are definitions?
2	CS 2, book pp. 399–419 (incl. the appendix)	Re-derive $b = \lim(1 + 1/n)^n$ from “ e^x is its own derivative” without looking	Where exactly does “maps sums to products” enter the derivative calculation?
3	CS 3, book pp. 420–432	Build $\mathbb{C}$ twice: once by rotation, once as $\mathbb{R}[x]/(x^2 + 1)$	Which “arbitrary” clauses of the formal construction does the rotation story <i>predict</i> , and which does it not?
4	CS 4, and above all pp. 446–451	Write out the four shared properties and check each one for both functions	Is “they entail the same properties” a <i>reason</i> for the identity, or a <i>restatement</i> of it?

Table 6. Session 4’s question is the one worth arguing with yourself about. My own answer is: a reason, but only under the weak reading of LEAK 6.

If you are short on time

CS 2 and CS 3 carry the most original content. CS 1 is setup you can skim once you have the three-stage figure. CS 4’s payoff section — “What Are the Ideas?”, book pp. 446–451 / pdf pp. 465–470 — can be read almost standalone once you have CS 1–3 in hand.

8 Appendix: complete inventory of Part VI

8.1 Every metaphor and blend, with its page

Name	book p.	pdf p.	What it buys
Case Study 1			
CARTESIAN PLANE BLEND	385–386	404–405	geometry $\leftrightarrow$ algebra; n -dimensional visualisation
FUNCTIONS ARE NUMBERS	386	405	arithmetic on functions; later, power series
UNIT CIRCLE BLEND (4 domains)	388–392	407–411	arcs acquire length; a, b acquire numbers
TRIGONOMETRY METAPHOR	393	412	angles become numbers; $\cos \pi = -1$
RECURRENCE IS CIRCULARITY	395	414	periodicity; the second meaning of π
TRIGONOMETRY BLEND	396	415	sine and cosine as <i>functions</i>
POLAR–TRIGONOMETRY METAPHOR	396	415	(r, θ) coordinates
Case Study 2			
MULTIPLICATION IS ADDITION	405	424	the exponential, as arithmetized metaphor
EXPONENTIAL BLEND	406	425	the log/exp one-to-one correspondence
TIME IS DISTANCE	407	426	step 1 of mathematicizing change
LOGICAL INDEPENDENCE IS GEOMETRICAL ORTHOGONALITY	407	426	the two axes
INSTANTANEOUS SPEED IS AVERAGE SPEED OVER AN INFINITELY SMALL INTERVAL	408	427	the derivative
CHANGE IS MOTION	408	427	rate of change in general
NUMBERS ARE WHOLE NUMBERS THAT ARE SUMS OF THEIR PARTS	412	431	numbers as sums
INFINITE SUMS ARE LIMITS OF SEQUENCES OF PARTIAL SUMS	412	431	BMI special case
POWER-SERIES BLEND (6 domains)	413	432	term-by-term differentiation
Case Study 3			
FUNDAMENTAL METONYMY OF ALGEBRA	421	440	solving for an unknown “number”
NUMBERS ARE POINTS ON A LINE	424	443	the Number-Line blend
MULTIPLICATION BY -1 IS ROTATION TO THE SYMMETRY POINT	424–425	443–444	negatives as rotation
MULTIPLICATION-ROTATION BLEND	425	444	arithmetizes R_{-1}
ROTATION-PLANE BLEND	426	445	vector addition, scalar multiplication
90° ROTATION METAPHOR	428–429	447–448	$n^2 = -1$, hence i ; the complex plane
Case Study 4			
TRIGONOMETRIC COMPLEX PLANE BLEND	437	456	cis; complex multiplication in polar form
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Table 8. Page correspondence for `Where_Mathematics_Come_From_by_G.Lakoff_R.Nunez_2001.pdf` (514 pages). Add **19** to a printed book page to get the PDF page.

$$e^{\pi i} + 1 = 0$$

It brings together e (change), π (recurrence), i (rotation), 1 and 0 (the two identities) — and it is true because, under the metaphors and blends of classical mathematics, **that is what those symbols mean.**